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C++ Numerics Work In Progress

Note: this is an early draft. It's known to be incomplet and incorrekt, and it has lots of bad fomattting.

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1 Foundations

[foundations]

1.1 Mission

[mission]

The mission of this document is to collect different numbers related proposals in one to make it simpler to discover issues and to simplify editors work of porting the papers to $\text{\LaTeX}$ (all the $\text{\LaTeX}$ sources are available at <https://github.com/ZaMaZaN4iK/numeric-papers-in-latex.git>).

1.2 Style of presentation

[intro.style]

Existing wording from the C++ working draft - included to provide context - is presented without decoration.

Entire clauses / subclauses / paragraphs incorporated from different numbers related proposals are presented in a distinct cyan color.

1.3 Scope

[intro.scope]

This document provides numbers and their options, but does not provide algorithms or special functions.

1.4 Design principles

[intro.design]

Follow mathematical behavior where feasible.

- Make unbounded numbers very simple to use.
- Make unavoidable overflow and rounding predictable and controllable.
 - Overflow on any operation is unmanageable.
 - A rational number with fixed-size fields needs to round.
- Avoid or hide aliasing effects.
- Prefer compile-time errors to run-time errors.
- Prefer safe defaults to efficient defaults.

Provide types that match the taxonomic needs.

- E.g. fixed-point, extended floating-point.
- Support construction of new types by library authors.
 - Expose common implementation abstractions.

Strive for efficiency.

- Use efficient function parameter passing.
- Use efficient representations.
- Give speed priority to dynamically common operations.
- Provide composite operations when efficient.
 - E.g. shift and add, multiply and add.
- Identify opportunities for new hardware.

- E.g. rounding right shift.

- Prefer run-time efficiency over compile-time efficiency.

Ease adoption and use.

- Provide a consistent vocabulary.

- Enable value conversion between all applicable types.

- Handle parameter aliasing within the implementations.

Ease extension.

- Provide a mechanism for conversion that does not require n^2 operations or coordination between independent developers.

- Most parts of the implementation should need only C++, so provide a machine abstraction layer.

- Expose sound "building-block" abstractions.

1.5 Number type taxonomy [intro.taxonomy]

Types may be categorized by the representation constancy between argument and result.

same The size of the type is the same between argument and result. Overflow is pervasive on all operations.

- invariant - All aspects of the representation are the same.

- invariant - Different fields within the representation may have different sizes.

adaptive The size of the result is statically known, but it may be different from (and generally larger than) the argument.

- limited - There is a maximum representation. If adaptation requires more than the maximum, the expression is ill-formed. Overflow in expressions is avoided until the maximum representation is reached.

- unlimited - There is no a priori maximum representation. Overflow in expressions is avoided.

dynamic All aspects of the representation are dynamic. Overflow in variables is avoided.

Types may also be categorized by the size specification, which is generally by the number of bits in the representation or by the number of digits required.

2 Operations

[intro.operations]

The various types will share many operations. To reduce surprise, these operations should have consistent names for consistent purposes.

Table 1 — Operations

operator	name	operation
arithmetic		
-	neg	negate
+	add	add
-	sub	subtract
*	mul	multiply
	rdiv	divide with rounding
	pdiv	divide with promotion
	quorem	quotient and remainder
	mod	modulo
	rem	remainder; different sign matching rules
scale/shift		
	ssu	static scale up
	ssd	static scale down with rounding
	dsu	dynamic scale up
	dsd	dynamic scale down with rounding
	lsh	left shift
	ash	arithmetic right shift
	rsh	logical right shift
	rtl	rotate left
	rtr	rotate right
composite		
	add2	add with two addends
	sub2	subtract with two subtrahends
	lshadd	left shift and add
	lshsub	left shift and sub
	muladd	multiply and add
	mulsub	multiply and subtract
	muladd2	multiply and add with two addends
	mulsub2	multiply and subtract with two subtrahends
bitwise		
	not	1's complement
	and	and
	xor	exclusive or
	ior	inclusive or
	dif	difference (a&~b)

3 Machine Abstraction Layer

[machine_layer]

The machine abstraction layer enables a mostly machine-independent implementation of this specification.

3.1 Overflow-Detecting Operations [overflow_detecting_ops]

The overflow-detecting functions return a boolean true when the operation overflows, and a boolean false when the operation does not overflow. Compilers may assume that a true result is rare. When the return is false, the function writes the operation result through the given pointer. When the return is true, the pointer is not used and no write occurs.

The following functions are available. Within these prototypes `T` and `C` are any integer type. However, `C` is useful only when it does not have values that `T` has.

```
bool overflow_cvt( C* result, T value );
bool overflow_neg( T* result, T value );
bool overflow_lsh( T* product, T multiplicand, int count );
bool overflow_add( T* summand, T augend, T addend );
bool overflow_sub( T* difference, T minuend, T subtrahend );
bool overflow_mul( T* product, T multiplicand, T multiplier );
```

3.2 Overflow-Handling Operations [overflow_handling_ops]

Overflow-handling operations require an overflow handling mode. We represent the mode in C++ as an enumeration:

```
enum class overflow {
    impossible, undefined, abort, exception,
    special,
    saturate, modulo_shifted
};
```

Within the definition of the following functions, we use a defining function, which we do not expect will be directly represented in C++. It is `T overflow(mode, T lower, T upper, U value)` where `U` either

- has a range that is not a subset of the range of `T` or
- is evaluated as a real number expression.

Many C++ conversions already reduce the range of a value, but they do not provide programmer control of that reduction. We can give programmers control.

```
template<typename T, typename U> T convert(overflow mode, U value)
```

Returns: `overflow(mode, numeric_limits<T>::min, numeric_limits<T>::max, value)`.

```
template<overflow mode, typename T, typename U> T convert(U value)
```

Returns: `convert(mode, value)`.

Being able to specify overflow from a range of values of the same type is also helpful.

```
template<typename T> T limit(overflow mode, T lower, T upper, T value)
```

Returns: overflow(mode, lower, upper, value).

```
template<overflow mode, typename T> T limit(T lower, T upper, T value)
```

Returns: limit(mode, lower, upper, value).

Common arguments can be elided with convenience functions.

```
template<typename T> T limit_nonnegative(overflow mode, T upper, T value)
```

Returns: limit(mode, 0, upper, value).

```
template<overflow mode, typename T> T limit_nonnegative(T upper, T value)
```

Returns: limit_nonnegative(mode, upper, value).

```
template<typename T> T limit_signed(overflow mode, T upper, T value)
```

Returns: limit(mode, -upper, upper, value).

```
template<overflow mode, typename T> T limit_signed(T upper, T value)
```

Returns: limit_signed(mode, upper, value).

Two's-complement numbers are a slight variant on the above.

```
template<typename T> T limit_twoscomp(overflow mode, T upper, T value)
```

Returns: limit(mode, -upper-1, upper, value).

```
template<overflow mode, typename T> T limit_twoscomp(T upper, T value)
```

Returns: limit_twoscomp(mode, upper, value).

For binary representations, we can also specify bits instead. While this specification may seem redundant, it enables faster implementations.

```
template<typename T> T limit_nonnegative_bits(overflow mode, T upper, T value)
```

Returns: overflow(mode, 0, $2^{\text{upper}} - 1$, value).

```
template<overflow mode, typename T> T limit_nonnegative_bits(T upper, T value)
```

Returns: limit_nonnegative_bits(mode, upper, value).

```
template<typename T> T limit_signed_bits(overflow mode, T upper, T value)
```

Returns: overflow(mode, $-(2^{\text{upper}} - 1)$, $2^{\text{upper}} - 1$, value).

```
template<overflow mode, typename T> T limit_signed_bits(T upper, T value)
```

Returns: limit_signed_bits(mode, upper, value).

```
template<typename T> T limit_twoscomp_bits(overflow mode, T upper, T value)
```

Returns: overflow(mode, -2^{upper} , $2^{\text{upper}} - 1$, value).

```
template<overflow mode, typename T> T limit_twoscomp_bits(T upper, T value)
```

Returns: limit_twoscomp_bits(mode, upper, value).

Embedding overflow detection within regular operations can lead to enhanced performance. In particular, left shift is a important candidate operation within fixed-point arithmetic.

```
template<typename T> T scale_up(overflow mode, T value, int count)
```

Returns: `overflow(mode, numeric_limits<T>::min, numeric_limits<T>::max, value*2count)`.

```
template<overflow mode, typename T> T scale_up(T value, int count)
```

Returns: `scale_up(mode, value, count)`.

3.3 Rounding Operations

[round_ops]

We represent the rounding mode in C++ as an enumeration:

```
enum class rounding {
    all_to_neg_inf, all_to_pos_inf,
    all_to_zero, all_away_zero,
    all_to_even, all_to_odd,
    all_fastest, all_smallest,
    all_unspecified,
    tie_to_neg_inf, tie_to_pos_inf,
    tie_to_zero, tie_away_zero,
    tie_to_even, tie_to_odd,
    tie_fastest, tie_smallest,
    tie_unspecified
};
```

The unmotivated modes `all_away_zero`, `all_to_even`, `all_to_odd`, `tie_to_neg_inf`, `tie_to_pos_inf`, and `tie_to_zero` are conditionally supported.

Within the definition of the following functions, we use a defining function, which we do not expect will be directly represented in C++. It is `T round(mode,U)` where `U` either

- has a finer resolution than `T` or
- is evaluated as a real number expression.

We already have rounding functions for converting floating-point numbers to integers. However, we need a facility that extends to different sizes of floating-point and between other numeric types.

```
template<typename T, typename U> T convert(rounding mode, U value)
```

Returns: `round(mode, U)`.

```
template<rounding mode, typename T, typename U> T convert(U value)
```

Returns: `round<T>(mode, U)`.

A division function has obvious utility.

```
template<typename T> T divide(rounding mode, T dividend, T divisor)
```

Returns: `is round(mode, dividend/divisor)`. Remember that division evaluates as a real number. Obviously, the implementation will use a different strategy, but it must yield the same result.

```
template<rounding mode, typename T> T divide(T dividend, T divisor)
```

Returns: `divide(mode, dividend, divisor)`.

Division by a power of two has substantial implementation efficiencies, and is used heavily in fixed-point arithmetic as a scaling mechanism. We represent the conjunction of these approaches with a rounding scale down (right shift).

```
template<typename T> T scale_down(rounding mode, T value, int bits)
```

Returns: round(mode, dividend/ 2^{bits}).

```
template<rounding mode, typename T> T scale_down(T value, int bits)
```

Returns: scale_down(mode, dividend, bits).

3.4 Combined Rounding and Overflow Operations [combined_rounding]

Some operations may reasonably both require rounding and require overflow detection.

First and foremost, conversion from floating-point to integer may require handling a floating-point value that has both a finer resolution and a larger range than the integer can handle. The problem generalizes to arbitrary numeric types.

```
template<typename T, typename U> T convert(overflow omode, rounding rmode, U value)
```

Returns: overflow(omode, numeric_limits<T>::min, numeric_limits<T>::max, round(rmode,value)).

```
template<overflow omode, rounding rmode, typename T, typename U> T convert(U value)
```

Returns: convert(omode, rmode; value).

Consider shifting as multiplication by a power of two. It has an analogy in a bidirectional shift, where a positive power is a left shift and a negative power is a right shift.

```
template<typename T> T scale(overflow omode, rounding rmode, T value, int count)
```

Returns: count < 0 ? round(rmode,value* 2^{count}) : overflow(omode, numeric_limits<T>::min, numeric_limits<T>::max, value* 2^{count}).

```
template<overflow omode, rounding rmode, typename T> T scale(T value, int count)
```

Returns: scale(omode, rmode value count).

3.5 Double-Word Operations [double_word_ops]

There are two classes of functions, those that provide a result in a single double-wide type and those that provide a result split into two single-wide types.

We expect programmers to use type names from `<cstdint>` or the parametric type aliases (below). Hence, we do not need to provide a means to infer one type size from the other. Within this section, we name these types as follows.

Table 2 — Double-word operations

name	description
S	signed integer type
U	unsigned integer type
DS	signed integer type that is double the width of the S type
DU	unsigned integer type that is double the width of the U type

We need a mechanism to specify the largest supported type for various combinations of function category and operation category. To that end, we propose macros as follows.

Table 3 — Macros

macro name	result category	operation category
LARGEST_DOUBLE_- WIDE_ADD	double-wide	add, add2, sub, sub2
LARGEST_DOUBLE_- WIDE_LSH	double-wide	lsh, lshadd
LARGEST_DOUBLE_- WIDE_MUL	double-wide	mul, muladd, muladd2, mulsub, mulsub2
LARGEST_DOUBLE_- WIDE_DIV	double-wide	divn, divw, divnrem, divwrem
LARGEST_DOUBLE_- WIDE_ALL	double-wide	the minimum size of the four macros above
LARGEST_SINGLE_- WIDE_ADD	single-wide	add, add2, sub, sub2
LARGEST_SINGLE_- WIDE_LSH	single-wide	lsh, lshadd
LARGEST_SINGLE_- WIDE_MUL	single-wide	mul, muladd, muladd2, mulsub, mulsub2
LARGEST_SINGLE_- WIDE_DIV	single-wide	divn, divw, divnrem, divwrem
LARGEST_SINGLE_- WIDE_ALL	double-wide	the minimum size of the four macros above

We need a mechanism to build and split double-wide types. The lower part of the split is always an unsigned type.

```
S split_upper( DS value );
U split_lower( DS value );
DS wide_build( S upper, U lower );
```

```
U split_upper( DU value );
U split_lower( DU value );
DU wide_build( U upper, U lower );
```

The arithmetic functions with an double-wide result are as follows. This category seems less important than the next category.

```
DS wide_lsh( S multiplicand, int count );
DS wide_add( S augend, S addend );
DS wide_sub( S minuend, S subtrahend );
DS wide_mul( S multiplicand, S multiplier );
DS wide_add2( S augend, S addend1, S addend2 );
DS wide_sub2( S minuend, S subtrahend1, S subtrahend2 );
DS wide_lshadd( S multiplicand, int count, S addend );
DS wide_lshsub( S multiplicand, int count, S subtrahend );
DS wide_muladd( S multiplicand, S multiplier, S addend );
DS wide_mulsub( S multiplicand, S multiplier, S subtrahend );
DS wide_muladd2( S multiplicand, S multiplier, S addend1, S addend2 );
DS wide_mulsub2( S multiplicand, S multiplier, S subtrahend1, S subtrahend2 );
```

```

S wide_divn( DS dividend, S divisor );
DS wide_divw( DS dividend, S divisor );
S wide_divnrem( S* remainder, DS dividend, S divisor );
DS wide_divnrem( S* remainder, DS dividend, S divisor );

DU wide_lsh( U multiplicand, int count );
DU wide_add( U augend, U addend );
DU wide_sub( U minuend, U subtrahend );
DU wide_mul( U multiplicand, U multiplier );
DU wide_add2( U augend, U addend1, U addend2 );
DU wide_sub2( U minuend, U subtrahend1, U subtrahend2 );
DU wide_lshadd( U multiplicand, int count, U addend );
DU wide_lshsub( U multiplicand, int count, U subtrahend );
DU wide_muladd( U multiplicand, U multiplier, U addend );
DU wide_mulsub( U multiplicand, U multiplier, U subtrahend );
DU wide_muladd2( U multiplicand, U multiplier, U addend1, U addend2 );
DU wide_mulsub2( U multiplicand, U multiplier, U subtrahend1, U subtrahend2 );
U wide_divn( DU dividend, U divisor );
DU wide_divw( DU dividend, U divisor );
U wide_divnrem( U* remainder, DU dividend, U divisor );
DU wide_divnrem( U* remainder, DU dividend, U divisor );

```

The arithmetic functions with a split result are as follows. The lower part of the result is always an unsigned type. The lower part is returned through a pointer while the upper part is returned as the function result. The intent is that in loops, the lower part is written once to memory while the upper part is carried between iterations in a local variable.

```

S split_lsh( U* product, S multiplicand, int count );
S split_add( U* summand, S augend, S addend );
S split_sub( U* difference, S minuend, S subtrahend );
S split_mul( U* product, S multiplicand, S multiplier );
S split_add2( U* summand, S value1, S addend1, S addend2 );
S split_sub2( U* difference, S minuend, S subtrahend1, S subtrahend2 );
S split_lshadd( U* product, S multiplicand, int count, S addend );
S split_lshsub( U* product, S multiplicand, int count, S subtrahend );
S split_muladd( U* product, S multiplicand, S addend1, S addend );
S split_mulsub( U* product, S multiplicand, S subtrahend1, S subtrahend2 );
S split_muladd2( U* product, S multiplicand, S multiplier, S addend1, S addend2 );
S split_mulsub2( U* product, S multiplicand, S multiplier, S subtrahend1, S subtrahend2 );
S split_divn( S dividend_upper, U dividend_lower, S divisor );
DS split_divw( S dividend_upper, U dividend_lower, S divisor );
S split_divnrem( S* remainder, S dividend_upper, U dividend_lower, S divisor );
DS split_divwrem( S* remainder, S dividend_upper, U dividend_lower, S divisor );

U split_lsh( U* product, U multiplicand, int count );
U split_add( U* summand, U value1, U addend );
U split_sub( U* difference, U minuend, U subtrahend );
U split_mul( U* product, U multiplicand, U multiplier );
U split_add2( U* summand, U value1, U addend1, U addend2 );
U split_sub2( U* difference, U minuend, U subtrahend1, U subtrahend2 );
U split_lshadd( U* product, U multiplicand, int count, U addend );
U split_lshsub( U* product, U multiplicand, int count, U subtrahend );
U split_muladd( U* product, U multiplicand, U multiplier, U addend );
U split_mulsub( U* product, U multiplicand, U multiplier, U subtrahend );
U split_muladd2( U* product, U multiplicand, U multiplier, U addend1, U addend2 );
U split_mulsub2( U* product, U multiplicand, U multiplier, U subtrahend1, U subtrahend2 );

```

```
U split_divn( U dividend_upper, U dividend_lower, U divisor );
DU split_divw( U dividend_upper, U dividend_lower, U divisor );
U split_divnrem( U* remainder, U dividend_upper, U dividend_lower, U divisor );
DU split_divwrem( U* remainder, U dividend_upper, U dividend_lower, U divisor );
```

4 Machine extension layer

[machine_ext_layer]

The machine extension layer enables the implementation of extended types.

4.1 Word-array operations

[word_array_ops]

A word is the type provided by `LARGEST_SINGLE_WIDE_ALL` and defined above.

We provide the following operations. These operations are not intended to provide complete multi-word operations, but rather to handle subarrays with uniform operations. Higher-level operations then compose these operations into a complete operation.

```
U unsigned_subarray_addin_word U* multiplicand, int length, U addend );
```

Effects: Add the word `addend` to the `multiplicand` of length `length`, leaving the result in the `multiplicand`.

Returns: Any carry out from the accumulator.

```
U unsigned_subarray_add_word U* summand, const U* augend, int length, U addend );
```

Effects: Add the `addend` to the `augend` of length `length` writing the result to the `summand`, which is also of length `length`.

Returns: Any carry out from the `summand`.

```
U unsigned_subarray_mulin_word U* product, int length, U multiplier );
```

Effects: Multiply the `product` of length `length` by the `multiplier`, leaving the result in the `product`.

Returns: Any carry out from the `product`.

```
U unsigned_subarray_mul_word U* product, U* multiplicand, int length, U multiplier );
```

Effects: Multiply the `multiplicand` of length `length` by the `multiplier` writing the result to the `product`, which is also of length `length`.

Returns: Any carry out from the `product`.

```
U unsigned_subarray_accmul_word U* accumulator, U* multiplicand, int length, U multiplier );
```

Effects: Multiply the `multiplicand` of length `length` by the `multiplier` adding the result to the `accumulator`, which is also of length `length`.

Returns: Any carry out from the `accumulator`.

For each of the two add operations above, there is a corresponding subtract operation.

For each of the seven operations above (add+sub+mul), there is a corresponding signed operation. The primary difference between the two is sign extension.

For each of the fourteen operations in above, there is a corresponding operation where the 'right-hand' argument is a pointer to a subarray, which is also of length `length`.

5 Feature test macros (Informative)

[feature.test]

- ¹ If an implementation supplies all of the conditionally-supported features specified in ??, `<wide_integer>` header in this document shall additionally define the `__cpp_lib_wide_integer` feature test macro.

Table 4 — Feature-test macro(s)

Macro name	Value
<code>__cpp_lib_wide_integer</code>	201909

6 Wide Integers

[wide__integer]

6.1 Class template `numeric_limits`

[numeric.limits]

Add the following sentence after the sentence "Specializations shall be provided for each arithmetic type, both floating-point and integer, including bool." (first sentence in fourth paragraph in [numeric.limits]):

Specializations shall be also provided for `wide_integer` type.

[*Note:* If there is a built-in integral type `Integral` that has the same signedness and width as `wide_integer<Bits, S>`, then `numeric_limits<wide_integer<Bits, S>>` specialized in the same way as `numeric_limits<Integral>` — *end note*]

6.2 Header `<wide_integer>` synopsis

[numeric.wide__integer.syn]

```
namespace std {

    // 26.???.2 class template wide_integer
    template<size_t Bits, typename S> class wide_integer;

    // 26.???.?? type traits specializations
    template<size_t Bits, typename S, size_t Bits2, typename S2>
        struct common_type<wide_integer<Bits, S>, wide_integer<Bits2, S2>>;

    template<size_t Bits, typename S, typename Arithmetic>
        struct common_type<wide_integer<Bits, S>, Arithmetic>;

    template<typename Arithmetic, size_t Bits, typename S>
        struct common_type<Arithmetic, wide_integer<Bits, S>>
        : common_type<wide_integer<Bits, S>, Arithmetic>
        ;

    // 26.???.?? unary operations
    template<size_t Bits, typename S>
        constexpr wide_integer<Bits, S> operator~(const wide_integer<Bits, S>& val) noexcept;
    template<size_t Bits, typename S>
        constexpr wide_integer<Bits, S> operator-(const wide_integer<Bits, S>& val) noexcept(is_unsigned_v<S>);
    template<size_t Bits, typename S>
        constexpr wide_integer<Bits, S> operator+(const wide_integer<Bits, S>& val) noexcept(is_unsigned_v<S>);

    // 26.???.?? binary operations
    template<size_t Bits, typename S, size_t Bits2, typename S2>
        common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
            constexpr operator*(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs);

    template<size_t Bits, typename S, size_t Bits2, typename S2>
        common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
            constexpr operator/(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs);

    template<size_t Bits, typename S, size_t Bits2, typename S2>
        common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
            constexpr operator+(const wide_integer<Bits, S>& lhs,
                               const wide_integer<Bits2, S2>& rhs) noexcept(is_unsigned_v<S>);
```



```

template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
    constexpr operator-(const wide_integer<Bits, S>& lhs,
                        const wide_integer<Bits2, S2>& rhs) noexcept(is_unsigned_v<S>);

template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
    constexpr operator%(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs);

template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
    constexpr operator&(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;

template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
    constexpr operator|(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;

template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
    constexpr operator^(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;

template<size_t Bits, typename S>
common_type_t<wide_integer<Bits, S>, size_t>
    constexpr operator<<(const wide_integer<Bits, S>& lhs, size_t rhs);

template<size_t Bits, typename S>
common_type_t<wide_integer<Bits, S>, size_t>
    constexpr operator>>(const wide_integer<Bits, S>& lhs, size_t rhs);

template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator<(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;

template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator>(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;

template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator<=(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;

template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator>=(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;

template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator==(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;

template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator!=(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;

// 26.???.?? numeric conversions
template<size_t Bits, typename S> std::string to_string(const wide_integer<Bits, S>& val);
template<size_t Bits, typename S> std::wstring to_wstring(const wide_integer<Bits, S>& val);

// 26.???.?? iostream specializations
template<class Char, class Traits, size_t Bits, typename S>
    basic_ostream<Char, Traits>& operator<<(basic_ostream<Char, Traits>& os,

```

```

        const wide_integer<Bits, S>& val);

template<class Char, class Traits, size_t Bits, typename S>
    basic_istream<Char, Traits>& operator>>(basic_istream<Char, Traits>& is,
        wide_integer<Bits, S>& val) noexcept;

// 26.??? hash support
template<class T> struct hash;
template<size_t Bits, typename S> struct hash<wide_integer<Bits, S>>;

template <size_t Bits, typename S>
    to_chars_result to_chars(char* first, char* last, const wide_integer<Bits, S>& value,
        int base = 10);

template <size_t Bits, typename S>
    from_chars_result from_chars(const char* first, const char* last, wide_integer<Bits, S>& value,
        int base = 10);

template <size_t Bits>
using wide_int = wide_integer<Bits, signed>;

template <size_t Bits>
using wide_uint = wide_integer<Bits, unsigned>

// optional literals
inline namespace literals {
inline namespace wide_int_literals {

constexpr wide_int<128> operator "" _int128(const char*);
constexpr wide_int<256> operator "" _int256(const char*);
constexpr wide_int<512> operator "" _int512(const char*);
constexpr wide_uint<128> operator "" _uint128(const char*);
constexpr wide_uint<256> operator "" _uint256(const char*);
constexpr wide_uint<512> operator "" _uint512(const char*);

} // namespace wide_int_literals
} // namespace literals

} // namespace std

```

The header `<wide_integer>` defines class template `wide_integer` and a set of operators for representing and manipulating integers of specified width.

Example:

```

using int128_t = wide_int<128>;
constexpr int128_t c = std::numeric_limits<int128_t>::min();
static_assert(c == 0x80000000000000000000000000000000_uint128);

int256_t a = 13;
a += 0xFF;
a *= 2.0;
a -= 12_int128;
assert(a > 0);

```

6.3 Template class `wide_integer` overview

[numeric.wide_integer.overview]

```

namespace std {
    template<size_t Bits, typename S>
    class wide_integer {
    public:
        // 26.??2.?? construct:
        constexpr wide_integer() noexcept = default;
        constexpr wide_integer(const wide_integer<Bits, S>& ) noexcept = default;
        template<typename Arithmetic> constexpr wide_integer(const Arithmetic& other) noexcept;
        template<size_t Bits2, typename S2> constexpr wide_integer(const wide_integer<Bits2, S2>& other) noexcept;

        // 26.??2.?? assignment:
        constexpr wide_integer<Bits, S>& operator=(const wide_integer<Bits, S>& ) noexcept = default;
        template<typename Arithmetic>
            constexpr wide_integer<Bits, S>& operator=(const Arithmetic& other) noexcept;
        template<size_t Bits2, typename S2>
            constexpr wide_integer<Bits, S>& operator=(const wide_integer<Bits2, S2>& other) noexcept;

        // 26.??2.?? compound assignment:
        template<typename Arithmetic>
            constexpr wide_integer<Bits, S>& operator+=(const Arithmetic&);
        template<size_t Bits2, typename S2>
            constexpr wide_integer<Bits, S>& operator+=(const wide_integer<Bits2, S2>&);

        template<typename Arithmetic>
            constexpr wide_integer<Bits, S>& operator/=(const Arithmetic&);
        template<size_t Bits2, typename S2>
            constexpr wide_integer<Bits, S>& operator/=(const wide_integer<Bits2, S2>&);

        template<typename Arithmetic>
            constexpr wide_integer<Bits, S>& operator+=(const Arithmetic&) noexcept(is_unsigned_v<S>);
        template<size_t Bits2, typename S2>
            constexpr wide_integer<Bits, S>& operator+=(const wide_integer<Bits2, S2>&) noexcept(is_unsigned_v<S>);

        template<typename Arithmetic>
            constexpr wide_integer<Bits, S>& operator-=(const Arithmetic&) noexcept(is_unsigned_v<S>);
        template<size_t Bits2, typename S2>
            constexpr wide_integer<Bits, S>& operator-=(const wide_integer<Bits2, S2>&) noexcept(is_unsigned_v<S>);

        template<typename Integral>
            constexpr wide_integer<Bits, S>& operator%=(const Integral&);
        template<size_t Bits2, typename S2>
            constexpr wide_integer<Bits, S>& operator%=(const wide_integer<Bits2, S2>&);

        template<typename Integral>
            constexpr wide_integer<Bits, S>& operator&=(const Integral&) noexcept;
        template<size_t Bits2, typename S2>
            constexpr wide_integer<Bits, S>& operator&=(const wide_integer<Bits2, S2>&) noexcept;

        template<typename Integral>
            constexpr wide_integer<Bits, S>& operator|=(const Integral&) noexcept;
        template<size_t Bits2, typename S2>
            constexpr wide_integer<Bits, S>& operator|=(const wide_integer<Bits2, S2>&) noexcept;

        template<typename Integral>

```

```

    constexpr wide_integer<Bits, S>& operator^=(const Integral&) noexcept;
template<size_t Bits2, typename S2>
    constexpr wide_integer<Bits, S>& operator^=(const wide_integer<Bits2, S2>&) noexcept;

template<typename Integral>
    constexpr wide_integer<Bits, S>& operator<<=(const Integral&);
template<size_t Bits2, typename S2>
    constexpr wide_integer<Bits, S>& operator<<=(const wide_integer<Bits2, S2>&);

template<typename Integral>
    constexpr wide_integer<Bits, S>& operator>>=(const Integral&) noexcept;
template<size_t Bits2, typename S2>
    constexpr wide_integer<Bits, S>& operator>>=(const wide_integer<Bits2, S2>&) noexcept;

constexpr wide_integer<Bits, S>& operator++() noexcept(is_unsigned_v<S>);
constexpr wide_integer<Bits, S> operator++(int) noexcept(is_unsigned_v<S>);
constexpr wide_integer<Bits, S>& operator--() noexcept(is_unsigned_v<S>);
constexpr wide_integer<Bits, S> operator--(int) noexcept(is_unsigned_v<S>);

// 26.???.2.?? observers:
template<typename Arithmetic> constexpr operator Arithmetic() const noexcept;
constexpr explicit operator bool() const noexcept;
private:
    byte data[Bits / CHAR_BITS]; // exposition only
};
} // namespace std

```

The class template `wide_integer<size_t Bits, typename S>` is a trivial standard layout class that behaves as an integer type of a compile time specified bitness.

Template parameter `Bits` specifies exact bits count to store the integer value. `Bits`

When size of `wide_integer` is equal to a size of builtin integral type then the alignment and layout of that `wide_integer` is equal to the alignment and layout of the builtin type.

Template parameter `S` specifies signedness of the stored integer value and is either signed or unsigned.

Implementations are permitted to add explicit conversion operators and explicit or implicit constructors for `Arithmetic` and for `Integral` types.

Example:

```

template <class Arithmetic>
[[deprecated("Implicit conversions to builtin arithmetic types are not safe!")]]
    constexpr operator Arithmetic() const noexcept;

explicit constexpr operator bool() const noexcept;
explicit constexpr operator int() const noexcept;
...

```

6.3.1 `wide_integer` constructors

[[numeric.wide_integer.cons](#)]

```
constexpr wide_integer() noexcept = default;
```

Effects: Constructs an object with undefined value.

```

template<typename Arithmetic>
    constexpr wide_integer(const Arithmetic& other) noexcept;

```

Effects: Constructs an object from `other` using the integral conversion rules [[conv.integral](#)].

```
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits2, S2>& other) noexcept;
```

Effects: Constructs an object from `other` using the integral conversion rules [conv.integral].

6.3.2 wide_integer assignments

[numeric.wide__integer.assign]

```
template<typename Arithmetic>
constexpr wide_integer<Bits, S>& operator=(const Arithmetic& other) noexcept;
```

Effects: Constructs an object from `other` using the integral conversion rules [conv.integral].

```
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator=(const wide_integer<Bits2, S2>& other) noexcept;
```

Effects: Constructs an object from `other` using the integral conversion rules [conv.integral].

6.3.3 wide_integer compound assignments

[numeric.wide__integer.cassign]

```
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator*=(const wide_integer<Bits2, S2>&);
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator/=(const wide_integer<Bits2, S2>&);
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator+=(const wide_integer<Bits2, S2>&) noexcept(is_unsigned_v<S>);
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator-=(const wide_integer<Bits2, S2>&) noexcept(is_unsigned_v<S>);
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator%=(const wide_integer<Bits2, S2>&);
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator&=(const wide_integer<Bits2, S2>&) noexcept;
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator|=(const wide_integer<Bits2, S2>&) noexcept;
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator^=(const wide_integer<Bits2, S2>&) noexcept;
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator<=(const wide_integer<Bits2, S2>&);
template<size_t Bits2, typename S2>
constexpr wide_integer<Bits, S>& operator>=(const wide_integer<Bits2, S2>&) noexcept;
constexpr wide_integer<Bits, S>& operator++() noexcept(is_unsigned_v<S>);
constexpr wide_integer<Bits, S> operator++(int) noexcept(is_unsigned_v<S>);
constexpr wide_integer<Bits, S>& operator--() noexcept(is_unsigned_v<S>);
constexpr wide_integer<Bits, S> operator--(int) noexcept(is_unsigned_v<S>);
```

Effects: Behavior of the above operators is similar to operators for built-in integral types.

```
template<typename Arithmetic>
constexpr wide_integer<Bits, S>& operator*=(const Arithmetic&);
template<typename Arithmetic>
constexpr wide_integer<Bits, S>& operator/=(const Arithmetic&);
template<typename Arithmetic>
constexpr wide_integer<Bits, S>& operator+=(const Arithmetic&) noexcept(is_unsigned_v<S>);
template<typename Arithmetic>
constexpr wide_integer<Bits, S>& operator-=(const Arithmetic&) noexcept(is_unsigned_v<S>);
```

Effects: As if an object `wi` of type `wide_integer<Bits, S>` was created from input value and the corresponding operator was called for `*this` and the `wi`.

```

template<typename Integral>
    constexpr wide_integer<Bits, S>& operator%=(const Integral&);
template<typename Integral>
    constexpr wide_integer<Bits, S>& operator&=(const Integral&) noexcept;
template<typename Integral>
    constexpr wide_integer<Bits, S>& operator|=(const Integral&) noexcept;
template<typename Integral>
    constexpr wide_integer<Bits, S>& operator^=(const Integral&) noexcept;
template<typename Integral>
    constexpr wide_integer<Bits, S>& operator<<=(const Integral&);
template<typename Integral>
    constexpr wide_integer<Bits, S>& operator>>=(const Integral&) noexcept;

```

Effects: As if an object `wi` of type `wide_integer<Bits, S>` was created from input value and the corresponding operator was called for `*this` and the `wi`.

6.3.4 `wide_integer` observers [numeric.wide_integer.observers]

```
template <typename Arithmetic> constexpr operator Arithmetic() const noexcept;
```

Returns: If `is_integral_v<Arithmetic>` then `Arithmetic` is constructed from `*this` using the integral conversion rules [conv.integral]. If `is_floating_point_v<Arithmetic>`, then `Arithmetic` is constructed from `*this` using the floating-integral conversion rules [conv.fpint]. Otherwise the operator shall not participate in overload resolution.

6.4 Specializations of `common_type` [numeric.wide_integer.traits.specializations]

```

template<size_t Bits, typename S, size_t Bits2, typename S2>
struct common_type<wide_integer<Bits, S>, wide_integer<Bits2, S2>> {
    using type = wide_integer<max(Bits, Bits2), see below>;
};

```

The signed template parameter indicated by this specialization is following:

- `(is_signed_v<S> && is_signed_v<S2> ? signed : unsigned)` if `Bits == Bits2`
- `S` if `Bits > Bits2`
- `S2` otherwise

[Note: `common_type` follows the usual arithmetic conversions design. - end note]

[Note: `common_type` attempts to follow the usual arithmetic conversions design here for interoperability between different numeric types. Following two specializations must be moved to a more generic place and enriched with usual arithmetic conversion rules for all the other numeric classes that specialize `std::numeric_limits`- end note]

```

template<size_t Bits, typename S, typename Arithmetic>
struct common_type<wide_integer<Bits, S>, Arithmetic> {
    using type = see below;
};

```

```

template<typename Arithmetic, size_t Bits, typename S>
struct common_type<Arithmetic, wide_integer<Bits, S>>
: common_type<wide_integer<Bits, S>, Arithmetic>;

```

The member typedef type is following:

- `Arithmetic` if `numeric_limits<Arithmetic>::is_integer` is false

```

— wide_integer<Bits, S> if sizeof(wide_integer<Bits, S>) > sizeof(Arithmetic)
— Arithmetic if sizeof(wide_integer<Bits, S>) < sizeof(Arithmetic)
— Arithmetic if sizeof(wide_integer<Bits, S>) == sizeof(Arithmetic) && is_signed_v<S>
— Arithmetic if sizeof(wide_integer<Bits, S>) == sizeof(Arithmetic) && numeric_limits<wide_
  integer<Bits, S>>::is_signed == numeric_limits<Arithmetic>::is_signed
— wide_integer<Bits, S> otherwise

```

6.5 Unary operators

[numeric.wide_integer.unary_ops]

```

template<size_t Bits, typename S>
constexpr wide_integer<Bits, S> operator~(const wide_integer<Bits, S>& val) noexcept;
Returns: value with inverted significant bits of val.

template<size_t Bits, typename S>
constexpr wide_integer<Bits, S> operator-(const wide_integer<Bits, S>& val) noexcept(is_unsigned_v<S>);
Returns: val *= -1 if S is true, otherwise the result is unspecified.

template<size_t Bits, typename S>
constexpr wide_integer<Bits, S> operator+(const wide_integer<Bits, S>& val) noexcept(is_unsigned_v<S>);
Returns: val.

```

6.6 Binary operators

[numeric.wide_integer.binary_ops]

In the function descriptions that follow, CT represents `common_type_t<A, B>`, where A and B are the types of the two arguments to the function.

```

template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
constexpr operator*(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs);
Returns: CT(lhs) *= rhs.

template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
constexpr operator/(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs);
Returns: CT(lhs) /= rhs.

template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
constexpr operator+(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs)
    noexcept(is_unsigned_v<S>);
Returns: CT(lhs) += rhs.

template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
constexpr operator-(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs)
    noexcept(is_unsigned_v<S>);
Returns: CT(lhs) -= rhs.

```

```
template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
constexpr operator%(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs);
```

Returns: CT(lhs) %= rhs.

```
template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
constexpr operator&(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;
```

Returns: CT(lhs) &= rhs.

```
template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
constexpr operator|(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;
```

Returns: CT(lhs) |= rhs.

```
template<size_t Bits, typename S, size_t Bits2, typename S2>
common_type_t<wide_integer<Bits, S>, wide_integer<Bits2, S2>>
constexpr operator^(const wide_integer<Bits, S>& lhs, const wide_integer<Bits2, S2>& rhs) noexcept;
```

Returns: CT(lhs) ^= rhs.

```
template<size_t Bits, typename S>
common_type_t<wide_integer<Bits, S>, size_t>
constexpr operator<<(const wide_integer<Bits, S>& lhs, size_t rhs);
```

Returns: CT(lhs) <<= rhs.

```
template<size_t Bits, typename S>
common_type_t<wide_integer<Bits, S>, size_t>
constexpr operator>>(const wide_integer<Bits, S>& lhs, size_t rhs) noexcept;
```

Returns: CT(lhs) >>= rhs.

```
template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator<(const wide_integer<Bits, S>& lhs,
                        const wide_integer<Bits2, S2>& rhs) noexcept;
```

Returns: true if value of CT(lhs) is less than the value of CT(rhs).

```
template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator>(const wide_integer<Bits, S>& lhs,
                        const wide_integer<Bits2, S2>& rhs) noexcept;
```

Returns: true if value of CT(lhs) is greater than the value of CT(rhs).

```
template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator<=(const wide_integer<Bits, S>& lhs,
                        const wide_integer<Bits2, S2>& rhs) noexcept;
```

Returns: true if value of CT(lhs) is equal or less than the value of CT(rhs).

```
template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator>=(const wide_integer<Bits, S>& lhs,
                        const wide_integer<Bits2, S2>& rhs) noexcept;
```

Returns: true if value of CT(lhs) is equal or greater than the value of CT(rhs).


```
template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator==(const wide_integer<Bits, S>& lhs,
                           const wide_integer<Bits2, S2>& rhs) noexcept;
```

Returns: true if significant bits of CT(lhs) and CT(rhs) are the same.

```
template<size_t Bits, typename S, size_t Bits2, typename S2>
constexpr bool operator!=(const wide_integer<Bits, S>& lhs,
                           const wide_integer<Bits2, S2>& rhs) noexcept;
```

Returns: !(CT(lhs) == CT(rhs)).

6.7 Numeric conversions

[numeric.wide_integer.conversions]

```
template<size_t Bits, typename S> std::string to_string(const wide_integer<Bits, S>& val);
template<size_t Bits, typename S> std::wstring to_wstring(const wide_integer<Bits, S>& val);
```

Returns: Each function returns an object holding the character representation of the value of its argument. All the significant bits of the argument are outputted as a signed decimal in the style [-]dddd.

```
template <size_t Bits, typename S>
to_chars_result to_chars(char* first, char* last, const wide_integer<Bits, S>& value,
                          int base = 10);
```

Behavior of `wide_integer` overload is subject to the usual rules of primitive numeric output conversion functions [utility.to.chars].

```
template <size_t Bits, typename S>
from_chars_result from_chars(const char* first, const char* last, wide_integer<Bits, S>& value,
                             int base = 10);
```

Behavior of `wide_integer` overload is subject to the usual rules of primitive numeric input conversion functions [utility.from.chars].

6.8 iostream specializations

[numeric.wide_integer.io]

```
template<class Char, class Traits, size_t Bits, typename S>
basic_ostream<Char, Traits>& operator<<(basic_ostream<Char, Traits>& os,
                                         const wide_integer<Bits, S>& val);
```

1 *Effects:* As if by: `os << to_string(val)`.

2 *Returns:* `os`.

```
template<class Char, class Traits, size_t Bits, typename S>
basic_istream<Char, Traits>& operator>>(basic_istream<Char, Traits>& is,
                                         wide_integer<Bits, S>& val);
```

3 *Effects:* Extracts a `wide_integer` that is represented as a decimal number in the `is`. If bad input is encountered, calls `is.setstate(ios_base::failbit)` (which may throw `ios::failure` ([iostate.flags])).

4 *Returns:* `is`.

6.9 Hash support

[numeric.wide_integer.hash]

```
template<size_t Bits, typename S> struct hash<wide_integer<Bits, S>>;
```

The specialization is enabled (20.14.14). If there is a built-in integral type `Integral` that has the same typename and width as `wide_integer<Bits, S>`, and `wi` is an object of type `wide_integer<Bits, S>`, then `hash<wide_integer<MachineWords, S>>()(wi) == hash<Integral>()(Integral(wi))`.

7 Rational math

[rational.math]

7.1 Class rational

[rational.class]

```

class rational {
public:
    rational() noexcept;

    rational(const rational& rat);
    rational(rational&& rat) noexcept;

    explicit rational(float num);
    explicit rational(double num);
    explicit rational(long double num);

    explicit rational(integer num);

    rational(integer num, integer den);

    ~rational() noexcept;

    rational& operator=(const rational& rat);
    rational& operator=(rational&& rat) noexcept;

    rational& operator=(integer num);
    rational& assign(integer num, integer den);

    void swap(rational& rhs) noexcept;

    rational normalize() const;

    integer numer() const;
    integer denom() const;

    explicit operator bool() const noexcept;

    rational& negate() noexcept;
    rational& invert() noexcept;

    rational& operator++();
    rational& operator--();

    rational operator++(int);
    rational operator--(int);

    rational& operator+=(const integer& rhs);
    rational& operator-=(const integer& rhs);
    rational& operator*=(const integer& rhs);
    rational& operator/=(const integer& rhs);

    rational& operator+=(const rational& rhs);
    rational& operator-=(const rational& rhs);

```

```

    rational& operator*=(const rational& rhs);
    rational& operator/=(const rational& rhs);
};

```

The numerator and denominator shall be stored internally in a `std::experimental::seminumeric::integer`.
`rational` member functions:

```
rational() noexcept;
```

Effects: Constructs a `rational` with a numerator equal to zero and a denominator equal to one.

```

rational(const rational& rat);
rational(rational&& rat);

```

Effects: Constructs a `rational` with a value of `rat`.

```
explicit rational(integer num);
```

Effects: Constructs a `rational` with a value of `num`.

```

explicit rational(float val);
explicit rational(double val);
explicit rational(long double val);

```

Effects: Constructs a `rational` with a value equal to `val`.

```
rational(integer num, integer den);
```

Requires: `den != 0` *Effects:* Constructs a `rational` given the specified numerator and denominator.

```
~rational() noexcept;
```

Effects: Destructs `*this`.

```

rational& operator=(const rational& rhs);
rational& operator=(rational&& rhs) noexcept;

```

Effects: Assigns `rhs` to `*this`.

Returns: `*this`.

```
rational& operator=(integer num);
```

Effects: Assigns `num` to `*this`.

Returns: `*this`.

```
rational& assign(integer num, integer den);
```

Requires: `den != 0`

Effects: Assigns the specified numerator and denominator to `*this`.

Returns: `*this`.

```
void swap(rational& rhs) noexcept;
```

Effects: Swaps `*this` and `rhs`.

```
rational normalize() const;
```

Returns: A `rational` equal to `*this`, but with the numerator and denominator having no common factor other than 1 and the denominator greater than 0. If the numerator is 0, the denominator shall be 1.

`integer numer() const;`

Returns: The (possibly not normalized) numerator by value.

`integer denom() const;`

Returns: The (possibly not normalized) denominator by value.

`explicit operator bool() const noexcept;`

Returns: As if `*this != 0`.

`rational& negate() noexcept;`

Effects: Changes the sign of `*this`.

Returns: `*this`.

`rational& invert() noexcept;`

Requires: the numerator is non-zero.

Effects: Swaps the numerator and denominator.

Returns: `*this`.

7.1.1 `rational` member operators:

[`rational.member.ops`]

`rational& operator++();`

Effects: Adds 1 to `*this` and stores the result in `*this`.

Returns: `*this`.

`rational& operator--();`

Effects: Subtracts 1 from `*this` and stores the result in `*this`.

Returns: `*this`.

`rational operator++(int);`

Effects: Adds 1 to `*this` and stores the result in `*this`. *Returns:* The value of `*this` before the addition.

`rational operator--(int);`

Effects: Subtracts 1 from `*this` and stores the result in `*this`.

Returns: The value of `*this` before the subtraction.

`rational& operator+=(const integer& rhs);`

Effects: Adds the integer value `rhs` to `*this` and stores the result in `*this`.

Returns: `*this`.

`rational& operator-=(const integer& rhs);`

Effects: Subtracts the integer value `rhs` from `*this` and stores the result in `*this`.

Returns: `*this`.

```
rational& operator*=(const integer& rhs);
```

Effects: Multiplies `*this` by the integer value `rhs` and stores the result in `*this`.

Returns: `*this`.

```
rational& operator/=(const integer& rhs);
```

Requires: `rhs != 0`.

Effects: Divides `*this` by the integer value `rhs` and stores the result in `*this`.

Returns: `*this`.

```
rational& operator+=(const rational& rhs);
```

Effects: Adds the rational value `rhs` to `*this` and stores the result in `*this`.

Returns: `*this`.

```
rational& operator-=(const rational& rhs);
```

Effects: Subtracts the rational value `rhs` from `*this` and stores the result in `*this`.

Returns: `*this`.

```
rational& operator*=(const rational& rhs);
```

Effects: Multiplies `*this` by the rational value `rhs` and stores the result in `*this`.

Returns: `*this`.

```
rational& operator/=(const rational& rhs);
```

Requires: `rhs != 0`.

Effects: Divides `*this` by the rational value `rhs` and stores the result in `*this`.

Returns: `*this`.

7.1.2 rational non-member operators:

[rational.ops]

```
rational operator+(const rational& val);
```

Returns: `rational(val)`.

```
rational operator-(const rational& val);
```

Returns: `rational(val).negate()`.

```
rational operator+(const rational& lhs, const rational& rhs);
```

Returns: `rational(lhs) += rhs`.

```
rational operator-(const rational& lhs, const rational& rhs);
```

Returns: `rational(lhs) -= rhs`.

```
rational operator*(const rational& lhs, const rational& rhs);
```

Returns: `rational(lhs) *= rhs`.

```
rational operator/(const rational& lhs, const rational& rhs);
```

Requires: `rhs != 0`.

Returns: `rational(lhs) /= rhs`.

```

rational operator+(const rational& lhs, const integer& rhs);
    Returns: rational(lhs) += rhs.

rational operator-(const rational& lhs, const integer& rhs);
    Returns: rational(lhs) -= rhs.

rational operator*(const rational& lhs, const integer& rhs);
    Returns: rational(lhs) *= rhs.

rational operator/(const rational& lhs, const integer& rhs);
    Requires: rhs != 0.
    Returns: rational(lhs) /= rhs.

rational operator+(const integer& lhs, const rational& rhs);
    Returns: rational(rhs) += lhs.

rational operator-(const integer& lhs, const rational& rhs);
    Returns: rational(rhs).negate() += lhs.

rational operator*(const integer& lhs, const rational& rhs);
    Returns: rational(rhs) *= lhs.

rational operator/(const integer& lhs, const rational& rhs);
    Requires: rhs != 0.
    Returns: rational(rhs).invert() *= lhs.

bool operator==(const rational& lhs, const rational& rhs);
    Returns: As if lhs.numer() * rhs.denom() == rhs.numer() * lhs.denom().

bool operator!=(const rational& lhs, const rational& rhs);
    Returns: !(lhs == rhs).

bool operator<(const rational& lhs, const rational& rhs);
    Returns: As if lhs.numer() * rhs.denom() < rhs.numer() * lhs.denom().

bool operator>(const rational& lhs, const rational& rhs);
    Returns: rhs < lhs.

bool operator<=(const rational& lhs, const rational& rhs);
    Returns: lhs < rhs || lhs == rhs.

bool operator>=(const rational& lhs, const rational& rhs);
    Returns: lhs > rhs || lhs == rhs.

bool operator==(const rational& lhs, const integer& rhs);
    Returns: As if after normalization lhs.numer() == rhs && lhs.denom() == 1.

```

```
bool operator!=(const rational& lhs, const integer& rhs);
```

Returns: `!(lhs == rhs)`.

```
bool operator<(const rational& lhs, const integer& rhs);
```

Returns: As if `lhs.numer() < rhs * lhs.denom()`.

```
bool operator>(const rational& lhs, const integer& rhs);
```

Returns: As if `lhs.numer() > rhs * lhs.denom()`.

```
bool operator<=(const rational& lhs, const integer& rhs);
```

Returns: `lhs < rhs || lhs == rhs`.

```
bool operator>=(const rational& lhs, const integer& rhs);
```

Returns: `lhs > rhs || lhs == rhs`.

```
bool operator==(const integer& lhs, const rational& rhs);
```

Returns: `rhs == lhs`.

```
bool operator!=(const integer& lhs, const rational& rhs);
```

Returns: `rhs != lhs`.

```
bool operator<(const integer& lhs, const rational& rhs);
```

Returns: `rhs > lhs`.

```
bool operator>(const integer& lhs, const rational& rhs);
```

Returns: `rhs < lhs`.

```
bool operator<=(const integer& lhs, const rational& rhs);
```

Returns: `rhs >= lhs`.

```
bool operator>=(const integer& lhs, const rational& rhs);
```

Returns: `rhs <= lhs`.

8 Parametric aliases [parametric_aliases]

Parametric aliases provide a machine-independent mechanism to specify the desired allocation size built-in or extended types.

The following wording changes are relative to N4527.

8.1 Freestanding implementations [compliance]

Add the following entry to table 16.

Table 5 — Macros

chapter	description	header
18.4+	Floating-point types	<csdfloat>

8.2 General [support.general]

Add the following entry to table 29.

Table 6 — Macros

chapter	description	header
18.4+	Floating-point types	<csdfloat>

8.3 Header <csdint> [csdint.syn]

Add the following entries to the synopsis before paragraph 1.

```
namespace std {
    template<int bits> alias exact_2int = implementation-defined;
    template<int bits> alias fast_2int = implementation-defined;
    template<int bits> alias least_2int = implementation-defined;
    template<int bits> alias exact_2uint = implementation-defined;
    template<int bits> alias fast_2uint = implementation-defined;
    template<int bits> alias least_2uint = implementation-defined;
}
#define MAX_BITS_2INT implementation-defined;
#define MAX_BITS_2UINT implementation-defined;
```

8.4 Parametric types [csdint.parametric]

Add a new section with the following paragraphs.

The aliases below are conditionally supported. The macro MAX_BITS_2INT shall give the largest integer size (in bits) supported by the aliases. [Note: All variants support the same sizes. —end note] If these aliases are not supported, the value shall be 0. Any parameter to the alias shall be in the range 1 to MAX_BITS_2INT.

```
template<int bits> alias exact_2int
```


The alias `exact_2int` refers to a built-in signed binary integer type of exactly `bits` bits. If there are two types of the same size, it refers to the type that is closest to `int` in promotion order. The type must represent negative values with two's-complement representation.

```
template<int bits> alias fast_2int
```

The alias `fast_2int` refers to the fastest built-in signed binary integer type of at least `bits` bits. If there are two types of the same size, it refers to the type that is closest to `int` in promotion order. The type must represent negative values with two's-complement representation.

```
template<int bits> alias least_2int
```

The alias `least_2int` refers to the smallest built-in signed binary integer type of at least `bits` bits. If there are two types of the same size, it refers to the type that is closest to `int` in promotion order. The type must represent negative values with two's-complement representation.

```
template<int bits> alias exact_2uint
```

The alias `exact_2uint` refers to a built-in unsigned binary integer type of exactly `bits` bits. If there are two types of the same size, it refers to the type that is closest to unsigned `int` in promotion order.

```
template<int bits> alias fast_2uint
```

The alias `fast_2uint` refers to the fastest built-in unsigned binary integer type of at least `bits` bits. If there are two types of the same size, it refers to the type that is closest to unsigned `int` in promotion order.

```
template<int bits> alias least_2uint
```

The alias `least_2uint` refers to the smallest built-in unsigned binary integer type of at least `bits` bits. If there are two types of the same size, it refers to the type that is closest to unsigned `int` in promotion order.

8.5 Floating-point types

[cstdfloat]

After section 18.4, add a new section. It has no direct contents.

8.6 Header <cstdfloat>

[cstdfloat.syn]

Add a new section.

```
namespace std {
    template<int bits> alias exact_2ieeefloat = implementation-defined;
    template<int bits> alias fast_2ieeefloat = implementation-defined;
    template<int bits> alias least_2ieeefloat = implementation-defined;
    template<int bits> alias exact_10ieeefloat = implementation-defined;
    template<int bits> alias fast_10ieeefloat = implementation-defined;
    template<int bits> alias least_10ieeefloat = implementation-defined;
}
#define MAX_BITS_2IEEEFLOAT implementation-defined;
#define MAX_BITS_10IEEEFLOAT implementation-defined;
```

8.7 Parametric types

[cstdfloat.parametric]

Add a new section with the following paragraphs.

The aliases below are conditionally supported. The macro `MAX_BITS_2IEEEFLOAT` shall give the largest binary floating-point size (in bits) supported by the aliases. [Note: All variants support the same sizes. —end note] If none of these aliases are supported, the value shall be 0. The parameter to the alias shall be in the range 1 to `MAX_BITS_2IEEEFLOAT`.

```
template<int bits> alias exact_2ieeefloat
```

The alias `exact_2ieeefloat` refers to a built-in binary floating-point type of exactly `bits` bits. If there are two types of the same size, it refers to the type that is closest to double in promotion order. The type must use IEEE representation.

```
template<int bits> alias fast_2ieeefloat
```

The alias `fast_2ieeefloat` refers to the fastest built-in binary floating-point type of at least `bits` bits. If there are two types of the same size, it refers to the type that is closest to double in promotion order. The type must use IEEE representation.

```
template<int bits> alias least_2ieeefloat
```

The alias `least_2ieeefloat` refers to the smallest built-in binary floating-point type of at least `bits` bits. If there are two types of the same size, it refers to the type that is closest to double in promotion order. The type must use IEEE representation.

The aliases below are conditionally supported. The macro `MAX_BITS_10IEEEFLOAT` shall give the largest decimal floating point size (in bits) supported by the aliases. [Note: All variants support the same sizes. —end note] If none of these aliases are supported, the value shall be 0. The parameter to the alias shall be in the range 1 to `MAX_BITS_10IEEEFLOAT`.

```
template<int bits> alias exact_10ieeefloat
```

The alias `exact_10ieeefloat` refers to a built-in decimal floating-point type of exactly `bits` bits. If there are two types of the same size, it refers to the type that is closest to double in promotion order. The type must use IEEE representation.

```
template<int bits> alias fast_10ieeefloat
```

The alias `fast_10ieeefloat` refers to the fastest built-in decimal floating-point type of at least `bits` bits. If there are two types of the same size, it refers to the type that is closest to double in promotion order. The type must use IEEE representation.

```
template<int bits> alias least_10ieeefloat
```

The alias `least_10ieeefloat` refers to the smallest built-in decimal floating-point type of at least `bits` bits. If there are two types of the same size, it refers to the type that is closest to double in promotion order. The type must use IEEE representation.

9 Unbounded Types [unbounded__types]

9.1 Class `integer_data_proxy` [numeric.integer_data_proxy.syn]

```
namespace std {
    class integer_data_proxy {

        // type names
        typedef unspecified data_type;
        typedef unspecified arithmetic_type;
        typedef unspecified uarithmetic_type;
        typedef unspecified iterator;
        typedef unspecified const_iterator;
        typedef unspecified reverse_iterator;
        typedef unspecified const_reverse_iterator;

        // constructors
        integer_data_proxy(const integer_data_proxy& rhs) = delete;
        integer_data_proxy(integer_data_proxy&& rhs);

        // assign
        integer_data_proxy& operator=(const integer_data_proxy& rhs) = delete;
        integer_data_proxy& operator=(integer_data_proxy&& rhs) = delete;

        // iterators
        iterator begin() noexcept;
        iterator end() noexcept;
        reverse_iterator rbegin() noexcept;
        reverse_iterator rend() noexcept;
        const_iterator cbegin() const noexcept;
        const_iterator cend() const noexcept;
        const_reverse_iterator crbegin() const noexcept;
        const_reverse_iterator crend() const noexcept;

        // element access
        data_type operator[](size_t pos) const;
        data_type& operator[](size_t pos);

        // capacity
        size_t size() const noexcept;
        size_t capacity() const noexcept;
        void reserve(size_t digits);
        void shrink_to_fit();
    };
} // namespace std
```

The class describes an object that can be used to examine and modify the internal representation of an object of type `integer`. This allows advanced users to portably implement algorithms that are not provided natively.

There can be only one `integer_data_proxy` object associated with a particular `integer` object at any given time; that object is obtained by calling the `get_data_proxy` member function on the `integer` object. The

resulting object can be moved but not copied.

```
typedef unspecified arithmetic_type;
```

The typedef defines a synonym for a signed arithmetic type that is large enough to hold the product of the largest values that the implementation will store in an object of type `data_type`.

```
iterator begin();
```

Returns: An iterator object such that the iterators `[begin(), end())` point to the internal data elements of the `integer` object.

```
size_t capacity() const noexcept;
```

Returns: The number of decimal digits that the `integer` object can represent without reallocating its internal storage.

```
const_iterator cbegin() const;
```

Returns: An iterator object such that the iterator range `[cbegin(), cend())` points to the internal data elements of the `integer` object.

```
const_iterator cend() const;
```

Returns: An iterator object such that the iterator range `[cbegin(), cend())` points to the internal data elements of the `integer` object.

```
typedef unspecified const_iterator;
```

The typedef defines a synonym for an iterator that can be used to access but not modify internal data elements of the `integer` object.

```
typedef unspecified const_reverse_iterator;
```

The typedef defines a synonym for a reverse iterator that can be used to access but not modify internal data elements of the `integer` object.

```
const_reverse_iterator crbegin() const;
```

Returns: The member function returns a reverse iterator object such that the iterator range `[crbegin(), crend())` points to the internal data elements of the `integer` object in reverse order.

```
const_reverse_iterator crend() const;
```

Returns: A reverse iterator object such that the iterator range `[crbegin(), crend())` points to the internal data elements of the `integer` object in reverse order.

```
typedef unspecified data_type;
```

The typedef defines a synonym for the type of the `integer` object's internal data elements.

```
iterator end();
```

Returns: An iterator object such that the iterator range `[begin(), end())` points to the internal data elements of the `integer` object.

```
integer_data_proxy(const integer_data_proxy&) = delete;
```

The copy constructor is deleted.

```
integer_data_proxy(integer_data_proxy&& rhs);
```

Effects: Copies the contents of `rhs` and leaves `rhs` in an unspecified valid state.

```
typedef unspecified iterator;
```

The typedef defines a synonym for an iterator that can be used to access internal data elements of the `integer` object.

```
integer& operator=(const integer_data_proxy&) = delete;
integer& operator=(integer_data_proxy&&) = delete;
```

The copy assignment and move assignment operators are deleted.

```
data_type operator[](size_t pos) const;
```

Returns: The value of the internal data element at index `pos`.

```
data_type& operator[](size_t pos);
```

Returns: A reference to the internal data element at index `pos`.

```
reverse_iterator rbegin();
```

Returns: A reverse iterator object such that the iterator range `[crbegin(), crend())` points to the internal data elements of the `integer` object in reverse order.

```
reverse_iterator rend();
```

Returns: A reverse iterator object such that the iterator range `[crbegin(), crend())` points to the internal data elements of the `integer` object in reverse order.

```
void reserve(size_t digits);
```

Effects: Ensures that `capacity() >= digits`.

```
typedef unspecified reverse_iterator;
```

The typedef defines a synonym for a reverse iterator that can be used to access internal data elements of the `integer` object.

```
void shrink_to_fit();
```

Effect on original feature: Is a non-binding request to reduce `capacity()` to hold the `integer` object's current stored value without wasted space.

```
size_t size() const;
```

Returns: `capacity()`.

```
typedef unspecified uarithmetric_type;
```

The typedef defines a synonym for an unsigned arithmetic type that is large enough to hold the product of the largest values that the implementation will store in an object of type `data_type`.

9.2 Bits

[numeric.bits.syn]

```

namespace std {

    // 26.???.?? binary operations
    bits operator&(const bits& lhs, const bits& rhs);
    bits operator|(const bits& lhs, const bits& rhs);
    bits operator^(const bits& lhs, const bits& rhs);

    // 26.???.?? istream specializations
    template<class CharT, class Traits>
        basic_ostream<CharT, Traits>& operator<<(basic_ostream<CharT, Traits>& os,
                                                const bits& val);

    template<class CharT, class Traits>
        basic_ostream<CharT, Traits>& operator>>(basic_ostream<CharT, Traits>& os,
                                                bits& val);

} // namespace std

namespace std {
    class bits {
    public:

        class reference;

        // constructors
        bits() noexcept;

        template <class Ty>
            bits(Ty rhs) noexcept;    // integral types only

        bits(initializer_list<uint_least32_t> list);

        template <class CharT, class Traits, class Alloc>
            explicit bits(const basic_string<CharT, Traits, Alloc>& str,
                        typename basic_string<CharT, Traits, Alloc>::size_t pos = 0,
                        typename basic_string<CharT, Traits, Alloc>::size_t count = std::basic_string<CharT>::npos,
                        CharT zero = CharT('0'),
                        CharT one = CharT('1'));

        template <class CharT>
            explicit bits(const CharT *ptr,
                        typename basic_string<CharT>::size_t count = std::basic_string<CharT>::npos,
                        CharT zero = CharT('0'),
                        CharT one = CharT('1'));

        explicit bits(const integer& val);
        explicit bits(integer&& val);

        bits(const bits& rhs);
        bits(bits&& rhs) noexcept;

        // assign and swap
        template <class Ty>
            bits& operator=(Ty rhs);    // integral types only
    };

```

```

bits& operator=(const integer& rhs);
bits& operator=(integer&& rhs);
bits& operator=(const bits& rhs);
bits& operator=(bits&& rhs);
void swap(bits& rhs) noexcept;

// conversions
unsigned long to_ulong() const;
unsigned long long to_ullong() const;
template <class CharT = char, class Traits = char_traits<CharT>, class Alloc = allocator<CharT>>
    basic_string<CharT, Traits, Alloc> to_string(CharT zero = CharT('0'), CharT one = CharT('1')) const;

// logical operations
bits& operator&=(const bits& rhs);
bits& operator|=(const bits& rhs);
bits& operator^=(const bits& rhs);
bits operator~() const;

bits& operator<=(size_t rhs);
bits& operator>=(size_t rhs);
bits& operator<<(size_t rhs) const;
bits& operator>>(size_t rhs) const;

// element access and modification
bits& set() noexcept;
bits& set(size_t pos, bool val = true);
bits& reset() noexcept;
bits& reset(size_t pos);
bits& flip() noexcept;
bits& flip(size_t pos);
bool operator[](size_t pos) const;
reference operator[](size_t pos);
bool test(size_t pos) const noexcept;
bool all() const noexcept;
bool any() const noexcept;
bool none() const noexcept;
size_t count() const noexcept;
size_t count_not_set() const noexcept;

// comparison
bool operator==(const bits& rhs) const noexcept;
bool operator!=(const bits& rhs) const noexcept;

// capacity
size_t size() const noexcept;
size_t capacity() const noexcept;
void reserve(size_t bit_count);
void shrink_to_fit();
};
} // namespace std

```

The class describes an object that represents an unbounded set of bits.

9.2.1 Constructors

[numeric.bits.cons]

```
bits() noexcept;
```

Effects: Constructs an object whose value is 0.

```
template <class Ty>
bits(Ty rhs) noexcept;    // integral types only
```

Effects: Constructs an object whose value is the ones-complement representation of `rhs`. Shall not take part in overload resolution unless the type `Ty` is an integral type.

```
bits(initializer_list<uint_least32_t> list);
```

Effects: Constructs an object whose stored value is equal to the elements of the `initializer_list` treated as a series of unsigned 32-bit digits with the leftmost digit being most significant. For example, the `initializer_list` `0xFE, 0xF0, 0xAA, 0x31` represents the value $0xFE * 323 + 0xF0 * 322 + 0xAA * 321 + 0x31 * 320$.

```
template <class CharT, class Traits, class Alloc>
explicit bits(const basic_string<CharT, Traits, Alloc>& str,
              typename basic_string<CharT, Traits, Alloc>::size_t pos = 0,
              typename basic_string<CharT, Traits, Alloc>::size_t count = basic_string<CharT>::npos,
              CharT zero = CharT('0'),
              CharT one = CharT('1'));

template <class CharT>
explicit bits(const CharT *ptr,
              typename basic_string<CharT>::size_t count = basic_string<CharT>::npos,
              CharT zero = CharT('0'),
              CharT one = CharT('1'));
```

Effects: Construct an object whose value is the value represented by their argument, treating zero as 0 and one as 1.

```
explicit bits(const integer& rhs);
explicit bits(integer&& rhs);
```

Effects: Construct objects whose value is the ones-complement representation of `rhs`.

```
bits(const bits& rhs);
bits(bits&& rhs) noexcept;
```

Effects: Construct objects with the same value as `rhs`. The move constructor leaves `rhs` in an unspecified valid state.

9.2.2 Operations

[numeric.bits.ops]

```
size_t capacity() const noexcept;
```

Returns: The number of bits that the object can represent without reallocating its internal storage.

```
size_t count() const noexcept;
```

Returns: The number of bits in `*this` that are set, or `static_cast<size_t>(-1)` if the number of bits that are set is too large to fit in an object of type `size_t`.

```
size_t count_not_set() const noexcept;
```

Returns: The number of bits in `*this` that are not set, or `static_cast<size_t>(-1)` if the number of bits that are not set is too large to fit in an object of type `size_t`.

```
void flip() const noexcept;
```


Effects: Toggles all the bits in the stored value.

`void flip(size_t pos);`

Effects: Toggles the bit at position `pos` in the stored value.

`bool none() const noexcept;`

Returns: True only if none of the bits in `*this` is set.

`void reserve(size_t bit_count);`

Effects: Ensures that `capacity() >= bit_count`.

`bits& reset() noexcept;`

Effects: Clears all the bits of `*this`.

Returns: `*this`.

`bits& reset(size_t pos);`

Effects: Clears the bit at position `pos`.

Returns: `*this`.

`void set() noexcept;`

Effects: Sets all the bits of `*this`.

Returns: `*this`.

`void set(size_t pos, bool val = true);`

Effects: Sets the bit at position `pos` in the stored value to `val`.

Returns: `*this`.

`void shrink_to_fit();`

Effects: Is a non-binding request to reduce `capacity()` to hold the current stored value without wasted space.

`size_t size() const noexcept;`

Returns: `capacity()`.

`bool test(size_t pos) const noexcept;`

Returns: True only if the bit at position `pos` in the stored value is non-zero.

9.2.3 Bits conversion

[numeric.bits.conv]

```
template <class CharT = char, class Traits = char_traits<CharT>, class Alloc = allocator<CharT>>
    basic_string<CharT, Traits, Alloc> to_string(CharT zero = CharT('0'),
                                                CharT one = CharT('1'));
```

Returns: A string representation of the bits in the value stored in `*this`, using `zero` to represent 0 and `one` to represent 1.

`unsigned long long to_ullong() const;`

Returns: A value equal to the stored value of `*this`. It throws an exception of type `range_error` if the value cannot be represented as an `unsigned long long`.

`unsigned long to_ulong() const;`

Returns: A value equal to the stored value of `*this`. It throws an exception of type `range_error` if the value cannot be represented as a `long long`.

9.2.4 Bits operators

[numeric.bits.operators]

```
template <class Ty>
    bits& operator=(Ty rhs);    // integral types only
```

Effects: Shall not take part in overload resolution unless the type `Ty` is an arithmetic type. The operator effectively executes `*this = integer(rhs)`.

Returns: `*this`.

```
bits& operator=(const bits& rhs);
bits& operator=(bits&& rhs);
```

Effects: Store the value of `rhs` into `*this`.

Returns: `*this`.

```
bits& operator=(const integer& rhs);
bits& operator=(integer&& rhs);
```

Effects: Store the ones-complement representation of `rhs` into `*this`.

Returns: `*this`.

```
bool operator==(const bits& rhs) const noexcept;
```

Returns: True only if the stored value in `*this` is the same as the stored value in `rhs`.

```
bool operator!=(const bits& rhs) const noexcept;
```

Returns: `!(*this == rhs)`.

```
bits operator&(const bits& lhs, const bits& rhs);
```

Returns: An object whose value is the bitwise AND of the values of `lhs` and `rhs`.

```
bits& operator&=(const bits& rhs);
```

Effects: Sets the value of `*this` to the bitwise AND of the values of `*this` and `rhs`.

Returns: A reference to `*this`.

```
bits operator|(const bits& lhs, const bits& rhs);
```

Returns: An object whose value is the bitwise inclusive OR of the values of `lhs` and `rhs`.

```
bits& operator|=(const bits& rhs);
```

Effects: Sets the value of `*this` to the bitwise inclusive OR of the values of `*this` and `rhs`.

Returns: `*this`.

```
bits operator^(const bits& lhs, const bits& rhs);
```

Returns: An object whose value is the bitwise exclusive OR of the values of `lhs` and `rhs`.

```
bits& operator^=(const bits& rhs);
```

Effects: Sets the value of `*this` to the bitwise exclusive OR of the values of `*this` and `rhs`.

Returns: `*this`.

```
bits operator~() const;
```

Returns: An object that holds the complement of the set of bits held by `*this`.

```
bits operator>>(const bits& lhs, size_t rhs);
```

Returns: An object whose stored value is the value of the bits in `lhs` shifted right `rhs` positions.

```
bits& operator>>=(size_t rhs);
```

Effects: Sets the stored value in `*this` to the value of the bits in `*this` shifted right `rhs` positions.

Returns: `*this`.

```
template <class CharT, class Traits>
    basic_istream<CharT, Traits>& operator>>(basic_istream<CharT, Traits>& is,
                                             bits& val);
```

Effects: Has the effect of `string temp; is >> temp; val = temp; .`

Returns: `is`.

```
bits operator<<(const bits& lhs, size_t rhs);
```

Returns: An object whose stored value is the value of the bits in `lhs` shifted left `rhs` positions.

```
bits& operator<<=(size_t rhs);
```

Effects: Sets the stored value in `*this` to the value of the bits in `*this` shifted left `rhs` positions.

Returns: `*this`.

```
template <class CharT, class Traits>
    basic_ostream<CharT, Traits>& operator<<(basic_ostream<CharT, Traits>& os,
                                             const bits& val);
```

Effects: Has the effect of `os << val.to_string()`.

Returns: `os`.

```
bool operator[](size_t pos) const;
```

Returns: The value of the bit at position `pos`.

```
reference operator[](size_t pos);
```

Returns: An object of type `bits::reference` that refers to the bit at position `pos`.

9.2.5 bits::reference class

[numeric.bits.reference]

```
namespace std {
    class bits {
        class reference {
        public:
            reference& operator=(bool val) noexcept;
            reference& operator=(const reference& rhs) noexcept;
            bool operator~() const noexcept;
            operator bool() const noexcept;
            reference& flip() noexcept;
        };
    };
} // namespace std
```

The nested class `bits::reference` describes an object that can be used to manage a particular bit in an object of type `bits`.

```
reference& flip() noexcept;
```

Effects: Toggles the bit that the object manages.

```
reference& operator=(bool rhs) noexcept;
```

Effects: Sets the bit that the object manages to the value of `rhs`.

```
reference& operator=(const reference& rhs) noexcept;
```

Effects: Sets the bit that the object manages to the value managed by `rhs`.

```
bool operator~() const noexcept;
```

Returns: True if the bit managed by the object is set, otherwise false.

```
operator bool() const noexcept;
```

Returns: True if the bit that the object manages is set.

9.3 Integer

[numeric.integer.syn]

```
namespace std {
    class integer {
    public:
        // constructors
        integer() noexcept;

        template <class Ty>
            integer(Ty rhs) noexcept; // arithmetic types only

        integer(initializer_list<uint_least32_t> init);

        template <class CharT, class Traits, class Alloc>
            explicit integer(const basic_string<CharT, Traits, Alloc>& str);

        explicit integer(const bits& rhs);
        explicit integer(bits&& rhs);

        integer(const integer& rhs);
        integer(integer&& rhs) noexcept;

        // assign and swap
        template <class Ty>
            integer& operator=(Ty rhs); // arithmetic types only
        integer& operator=(const bits& rhs);
        integer& operator=(bits&& rhs);
        integer& operator=(const integer& rhs);
        integer& operator=(integer&& rhs);
        void swap(integer& rhs) noexcept;

        // conversions
        explicit operator long long() const;
        explicit operator unsigned long long() const;
        explicit operator long double() const noexcept;
        explicit operator bool() const noexcept;

        // comparisons
```

```

    int compare(const integer& rhs) const noexcept;

    // arithmetic operations
    integer& operator+=(const integer& rhs);
    integer& operator-=(const integer& rhs);
    integer& operator*=(const integer& rhs);
    integer& operator/=(const integer& rhs);
    integer& operator%=(const integer& rhs);

    integer& operator++();
    integer operator++(int);
    integer& operator--();
    integer operator--(int);

    integer div(const integer& rhs);

    integer& abs() noexcept;
    integer& negate() noexcept;
    integer operator+() const noexcept;
    integer operator-() const noexcept;

    integer& operator<=(size_t rhs);
    integer& operator>=(size_t rhs);

    // numeric operations
    integer& sqr();
    integer& sqrt();
    integer& pow(const integer& exp);
    integer& mod(const integer& rhs);
    integer& mulmod(const integer& rhs, const integer& m);
    integer& powmod(const integer& exp, const integer& m);

    // observers
    bool is_zero() const noexcept;
    bool is_odd() const noexcept;

    // accessors
    integer_data_proxy get_data_proxy();

    // capacity
    size_t size() const noexcept;
    size_t capacity() const noexcept;
    void reserve(size_t digits);
    void shrink_to_fit();
};
} // namespace std

void swap(integer& lhs, integer& rhs) noexcept;

// comparisons
bool operator==(const integer& lhs, const integer& rhs) noexcept;
bool operator!=(const integer& lhs, const integer& rhs) noexcept;
bool operator<(const integer& lhs, const integer& rhs) noexcept;
bool operator<=(const integer& lhs, const integer& rhs) noexcept;
bool operator>(const integer& lhs, const integer& rhs) noexcept;

```

```

bool operator>=(const integer& lhs, const integer& rhs) noexcept;

// arithmetic operations
integer operator+(const integer& lhs, const integer& rhs);
integer operator-(const integer& lhs, const integer& rhs);
integer operator*(const integer& lhs, const integer& rhs);
integer operator/(const integer& lhs, const integer& rhs);
integer operator%(const integer& lhs, const integer& rhs);

pair<integer, integer> div(const integer& lhs, const integer& rhs);

integer abs(const integer& val);

integer operator<<(const integer& lhs, size_t rhs);
integer operator>>(const integer& lhs, size_t rhs);

// numeric operations
integer sqr(const integer& val);
integer sqrt(const integer& val);
integer pow(const integer& val, const integer& exp);
integer mod(const integer& lhs, const integer& rhs);
integer mulmod(const integer& lhs, const integer& rhs, const integer& m);
integer powmod(const integer& lhs, const integer& rhs, const integer& m);

integer gcd(const integer& a, const integer& b);
integer lcm(const integer& a, const integer& b);

// conversions
string to_string(const integer& val, int radix = 10);

// I/O operations
template <class CharT, class Traits>
    basic_ostream<CharT, Traits>& operator<<(basic_ostream<CharT, Traits>& str,
                                           const integer& val);

template <class CharT, class Traits>
    basic_istream<CharT, Traits>& operator>>(basic_istream<CharT, Traits>& str,
                                           integer& val);

```

The class describes an object that manages an unbounded-precision signed integral type that can be used in most contexts where an `int` could be used.

Any function specified to return an object of type `integer` may return an object of another type, provided all the `const` member functions of the class `integer` are also applicable to that type.

```
integer abs(const integer& other);
```

Returns: An object that holds the absolute value of `other`.

```
integer& abs() noexcept;
```

Effects: Sets the stored value of `*this` to its absolute value and returns `*this`.

```
size_t capacity() const noexcept;
```

Returns: The number of decimal digits that the object can represent without reallocating its internal storage.

```
int compare(const integer& rhs) const noexcept;
```

Returns: A value less than 0 if `*this` is less than `rhs`, 0 if `*this` is equal to `rhs`, and greater than 0 if `*this` is greater than `rhs`.

```
pair<integer, integer> div(const integer& lhs, const integer& rhs);
```

Returns: An object that is an instantiation of `pair`; its first field holds the quotient, `lhs / rhs`, and its second field holds the remainder, `lhs % rhs`.

```
integer div(const integer& rhs) const;
```

Returns: The remainder, `*this % rhs`, and stores the quotient, `*this / rhs`, into `*this`.

```
integer gcd(const integer& a, const integer& b);
```

Returns: An object whose value is the greatest common denominator of `a` and `b`.

```
integer_data_proxy get_data_proxy();
```

Returns: An object of type `integer_data_proxy` that can be used to examine and modify the internal storage of `*this`. If an object of type `integer_data_proxy` that refers to `*this` exists at the time of a call to this function, the function throws an exception object of type `runtime_error`.

9.3.1 Constructors

[[numeric.integer.ctors](#)]

```
integer() noexcept;
```

Effects: Constructs an object whose value is 0.

```
template <class Ty>
```

```
integer(Ty val) noexcept; // arithmetic types only
```

Effects: For integral types the constructor constructs an object whose value is `val`. For floating-point types the constructor constructs an object whose value is the value of `val` with any fractional part discarded. Shall not take part in overload resolution unless the type `Ty` is an arithmetic type.

```
integer(initializer_list<unspecified> list);
```

Effects: Constructs an object whose stored value is equal to the elements of the `initializer_list` treated as a series of unsigned 32-bit digits with the leftmost digit being most significant. For example, the `initializer_list` `0xFE, 0xF0, 0xAA, 0x31` represents the value `0xFE * 323 + 0xF0 * 322 + 0xAA * 321 + 0x31 * 320`.

```
template<class CharT, class Traits, class Alloc>
```

```
explicit integer(const basic_string<CharT, Traits, Alloc>& str);
```

Effects: Constructs an object whose value is the value represented by the `string` object. The `string` object shall have the form required for the `string` argument to the function `strtol` with a radix of base, and shall be interpreted as if by `strtol(str.c_str(), 0, base)`, except that the resulting value can never be outside the range of representable values.

```
integer(const bits& rhs);
```

```
integer(bits&& rhs);
```

Effects: Construct an object whose stored value is the value in the bit pattern in `rhs` interpreted as a ones-complement representation of an integer value.

```
integer(const integer& rhs);
```

```
integer(integer&& rhs) noexcept;
```

Effects: Construct objects with the same value as `rhs`. The move constructor leaves `rhs` in an unspecified valid state.

9.3.2 Operations

[numeric.integer.ops]

```
bool is_odd() const noexcept;
```

Returns: True only if the stored value represents an odd number.

```
bool is_zero() const noexcept;
```

Returns: True only if the stored value is zero.

```
integer lcm(const integer& a, const integer& b);
```

Returns: An object whose value is the least common multiple of a and b.

```
integer mod(const integer& lhs, const integer& rhs);
```

Returns: An object whose value is lhs mod rhs.

```
integer& mod(const integer& rhs);
```

Effects: Sets the stored value in *this to *this mod rhs.

Returns: *this.

```
integer mulmod(const integer& lhs, const integer& rhs, const integer& m);
```

Returns: An object whose value is (lhs * rhs) mod m.

```
integer& mulmod(const integer& rhs, const integer& m);
```

Effects: Sets the value of *this to (*this * rhs) mod m.

Returns: *this.

```
integer& negate() noexcept;
```

Effects: Sets the stored value of *this to the negation of its previous value.

Returns: this.

```
integer pow(const integer& val, const integer& exp);
```

Requires: 0 <= exp.

Returns: An object whose value is val^{exp} .

```
integer& pow(const integer& exp);
```

Requires: 0 <= exp.

Effects: Sets the value of *this to $*this^{exp}$.

Returns: *this.

```
integer powmod(const integer& val, const integer& exp, const integer& m);
```

Requires: 0 <= exp and m != 0.

Returns: An object whose value is $val^{exp} \bmod m$.

```
integer& powmod(const integer& exp, const integer& m);
```

Requires: 0 <= exp and m != 0.

Effects: Sets the value of *this to $*this^{exp} \bmod m$.

Returns: *this.


```
void reserve(size_t digits);
```

Effects: Ensures that `capacity()` $\geq$ `digits`.

```
void shrink_to_fit();
```

Effects: A non-binding request to reduce `capacity()` to hold the current stored value without wasted space.

```
size_t size() const noexcept;
```

Returns: `capacity()`.

```
integer sqr(const integer& val);
```

Returns: An object whose value is `val * val`.

```
integer& sqr();
```

Effects: Sets the value of `*this` to `*this * *this`.

Returns: `*this`.

```
integer sqrt(const integer& val);
```

Requires: $0 \leq \text{val}$.

Returns: An object whose value is the square root of the value held by `val`, discarding any fractional part.

```
integer& sqrt();
```

Requires: $0 \leq *this$.

Effects: Sets the value of `*this` to the square root of the value held by `*this`, discarding any fractional part.

Returns: `*this`.

```
void swap(integer& lhs, integer& rhs) noexcept;
```

Effects: Swaps the stored values of `lhs` and `rhs`.

```
void swap(integer& rhs) noexcept;
```

Effects: Swaps the stored values of `*this` and `rhs`.

```
string to_string(const integer& val, int radix = 10) const;
```

Returns: A `string` representation of the value stored in `val`, using `radix` as the radix.

9.3.3 Conversion

[[numeric.integer.conv](#)]

```
explicit operator bool() const noexcept;
```

Returns: False only if `*this` is equal to 0.

```
explicit operator long double() const noexcept;
```

Returns: A value equal to the stored value of `*this`. If the stored value is outside the range that can be represented by an object of type `long double` the returned value is positive or negative infinity, as appropriate.

```
explicit operator long long() const;
```

Returns: A value equal to the stored value of `*this`. If the stored value cannot be represented as a `long long` it throws an exception of type `range_error`.

```
explicit operator unsigned long long() const;
```

Returns: A value equal to the stored value of `*this`. If the stored value cannot be represented as an `unsigned long long` it throws an exception of type `range_error`.

9.3.4 Comparison

[numeric.integer.comp]

```
bool operator==(const integer& lhs, const integer& rhs) noexcept;
```

Returns: True only if the value stored in `lhs` is equal to the value stored in `rhs`.

```
bool operator!=(const integer& lhs, const integer& rhs) noexcept;
```

Returns: `!(lhs == rhs)`.

```
bool operator>(const integer& lhs, const integer& rhs) noexcept;
```

Returns: `rhs < lhs`.

```
bool operator>=(const integer& lhs, const integer& rhs) noexcept;
```

Returns: `!(lhs < rhs)`.

```
bool operator<(const integer& lhs, const integer& rhs) noexcept;
```

Returns: True only if `lhs.compare(rhs)` returns -1.

```
bool operator<=(const integer& lhs, const integer& rhs) noexcept;
```

Returns: `!(rhs < lhs)`.

9.3.5 Assignment

[numeric.integer.assign]

```
template <class Ty>
integer& operator=(Ty rhs);    // arithmetic types only
```

Effects: Shall not take part in overload resolution unless the type `Ty` is an arithmetic type. The operator effectively executes `*this = integer(rhs)`.

Returns: `*this`.

```
integer& operator=(const integer& rhs);
integer& operator=(integer&& rhs);
```

Effects: Store the value of `rhs` into `*this`.

Returns: `*this`.

```
integer& operator=(const bits& rhs);
integer& operator=(bits&& rhs);
```

Effects: Store the value of `rhs`, interpreted as a ones-complement representation of an integer value, into `*this`.

Returns: `*this`.

9.3.6 Arithmetic operations

[numeric.integer.arithmetic_ops]

```
integer operator+(const integer& lhs, const integer& rhs);
```

Returns: An object whose value is the sum of the values of `lhs` and `rhs`.

```
integer operator+() const noexcept;
```

Returns: A copy of `*this`.

```
integer& operator+=(const integer& rhs);
```

Effects: Sets the stored value of `*this` to the sum of the values of `*this` and `rhs`.

Returns: A reference to `*this`.

```
integer& operator++();
```

Effects: Set the value stored in `*this` to `*this + 1`.

Returns: `*this`.

```
integer operator++(int);
```

Returns: An object whose value is the value stored in `*this` prior to the increment.

```
integer operator-(const integer& lhs, const integer& rhs)
```

Returns: An object whose value is the difference between the values of `lhs` and `rhs`.

```
integer operator-() noexcept;
```

Returns: An object whose value is the negation of the value of `*this`.

```
integer& operator--(const integer&);
```

Effects: Sets the stored value of `*this` to the difference between the values of `*this` and `rhs`.

Returns: `*this`.

```
integer& operator--();
```

Effects: Set the value stored in `*this` to `*this - 1`.

Returns: `*this`.

```
integer operator--(int);
```

Effects: Set the value stored in `*this` to `*this - 1`.

Returns: An object whose value is the value stored in `*this` prior to the decrement.

```
integer operator*(const integer& lhs, const integer& rhs);
```

Returns: An object whose value is the product of the values of `lhs` and `rhs`.

```
integer& operator*=(const integer& rhs);
```

Effects: Sets the stored value of `*this` to the product of the values of `*this` and `rhs`.

Returns: A reference to `*this`.

```
integer operator/(const integer& lhs, const integer& rhs);
```

Returns: An object whose value is the quotient of the value of `lhs` divided by the value of `rhs`, discarding any fractional part.

```
integer& operator/=(const integer& rhs);
```

Effects: Sets the stored value of `*this` to the quotient of the value of `*this` divided by the value of `rhs`, discarding any fractional part.

Returns: `*this`.

```
integer operator%(const integer&, const integer&);
```

Returns: An object whose value is the remainder of the value of `lhs` divided by the value of `rhs`. The remainder is the value such that $(lhs / rhs) * rhs + lhs \% rhs$ is equal to `lhs`.

```
integer& integer::operator%=(const integer&);
```

Effects: Sets the stored value of `*this` to the remainder of `*this` divided by the value of `rhs`.

Returns: `*this`.

```
integer operator>>(const integer& val, size_t rhs);
```

Returns: An object whose value is $val / 2^{rhs}$.

```
integer& operator>>=(size_t rhs);
```

Effects: Sets the value of `*this` to $*this / 2^{rhs}$.

Returns: `*this`.

9.3.7 I/O

[numeric.integer.io]

```
template <class Elem, class Traits>
```

```
    basic_istream<Elem, Traits>& operator>>(basic_istream<Elem, Traits>& is, integer& val);
```

Effects: Has the effect of `string temp; is >> temp; val = integer(temp);` .

Returns: `is`.

```
integer operator<<(const integer& val, size_t rhs);
```

Returns: An object whose value is $val * 2^{rhs}$.

```
integer& integer::operator<<=(size_t rhs);
```

Effects: Sets the value of `*this` to $*this * 2^{rhs}$.

Returns: `*this`.

```
template <class Elem, class Traits>
```

```
    basic_ostream<Elem, Traits>& operator<<(basic_ostream<Elem, Traits>& os, const integer& val);
```

Effects: Has the effect of `os << to_string(val)`.

Returns: `os`.

10 Generalized Type Conversion

[generalized_type_conversion]

Conversion between arbitrary numeric types requires something more practical than implementing the full cross product of conversion possibilities.

To that end, we propose that each numeric type promotes to an unbound type in its same general category. For example, integers of fixed size would promote to an unbound integer. In this promotion, there can be no possibility of overflow or rounding. Each type also demotes from that type. The demotion may have both round and overflow.

The general template conversion algorithm from type S to type T is to:

- Promote S to its unbound type S' .
- Convert S' to unbound type T' of T .
- Demote T' to T .

We expect common conversions to have specialized implementations.